Checkpoint 3a. Target architecture

Project: Fittybird x Kiabi, 360° Framework
Checkpoint: 3a, Period 3 (technical building blocks)
Terms: `shared/glossary.docx`

The architecture is organized in the four categories the framework asks for: **Systems**, **Techniques**, **Application** and **Functional**. Each category answers a different question, and every element is classified in exactly one of them.

Category	The question it answers	What goes in it
Functional	What does the product do for its users?	Business capabilities and the user-facing functions
Application	Which software pieces deliver those functions?	Components, interfaces, clients
Techniques	Which methods and technologies build those pieces?	Languages, frameworks, models, protocols, design rules
Systems	Where does it run, and how is it operated?	Hosting, networks, environments, pipelines, operations, Kiabi's own systems



Status words used below: **Decided** (the team has chosen), **Candidate** (the team's working choice, to confirm), **Open** (a decision is still to be made, with the sprint that settles it).

1. Functional architecture

Function	What it does	Backlog	Persona
Consent and account	Records consent, signs the shopper in with an account or the loyalty card, deletes data on request	US-01, US-02, US-03	Nadia
Create avatar	Guides photo capture, builds the avatar, confirms height and usual size	US-04, US-05, US-06, US-11	Léa
Browse catalog	Shows Kiabi articles with filters	US-07	Lucas
Try on and read fit	Shows an article on the avatar and rates each body zone	US-08, US-09, US-10, US-16	Léa
Buy through Kiabi cart	Hands the chosen article and size to Kiabi's cart and checkout	US-12	Lucas
In-store session	Loads the avatar on the mirror from a card tap or QR code, sends size	US-13, US-14, US-17	Karim,

	requests to a staff tablet		Inès
Feedback and measurement	Captures fit feedback and return reasons, records events	US-15, US-22	Claire
Insight for Kiabi	Shows usage, size choices and cross-channel sessions	US-18	Claire

2. Application architecture

Component	Role	Talks to	Status
App and web client	Shopper interface on phone and browser	API	Decided
Mirror and tablet client	In-store screen and staff tablet. The same web application in a different layout	API	Candidate
Dashboard client	Interface for the Kiabi team	API	Candidate
API and sign-in	Single entry point for every client. Handles authentication and routing	All services	Decided
Avatar service	Turns photos into an avatar	API, storage	Open (SP-01, sprint 1)
Fit engine	Compares the avatar with a size chart and returns the rating per zone	Avatar service, Kiabi adapter	Decided
Kiabi adapter	The only door to Kiabi's systems: catalog, size charts, cart hand-off, loyalty identity	Kiabi systems	Decided
Consent and storage	Stores avatars, consent log and profile data	API	Decided
Event analytics	Receives usage and feedback events and serves aggregates to the dashboard	API	Candidate

3. Techniques

Area	Choice	Why	Status
Client	Web application framework, responsive from phone to tablet to mirror	One codebase for three surfaces. Framework to be chosen by the UI lead in sprint 1 (React is the working assumption)	Candidate
API	REST with JSON, token-based sign-in	Simple, testable, easy to hand to a development team	Candidate
Photo to avatar	Build or buy. Compare three options on the same 10 sample photo sets (quality, cost, speed, privacy)	The largest technical unknown (risks R6, R7)	**Open**, decided at the end of sprint 1 (D4)
Fit calculation	Rule-based comparison between avatar measurements and the size chart, shown as a range with a size confirmation step	Explainable, needs no training data, protects trust (risk R3)	Decided (D7)
Data minimization	Photos deleted once the avatar is made unless the shopper opts in to keep them	Answers risks R1 and R9	Decided (D5)
Consent	Versioned consent log with date	Proves what the shopper agreed to	Decided
Identity in store	QR code stands in for the NFC tap in the first pilot	Removes the integration risk R10 from the critical path	Decided (D6)
Quality	Automated tests on the API and the fit engine, code review by a second person	Backs the Definition of Done	Candidate
Measurement	Event schema defined before the first pilot, with a control group	Needed to prove the return-rate effect	Decided (US-22)

4. Systems

Area	Choice	Why	Status
Hosting	A cloud region in the European Union	Personal data of French shoppers	Decided. Provider to choose in sprint 1

Data stores	A relational database for profiles and consent, object storage for avatars	Different data, different retention	Candidate
Environments	Four: Local, Dev, Staging, Production (see section 6)	Safe path from laptop to pilot	Decided
Delivery pipeline	Automated build, test and deploy on every merge	Keeps staging current for reviews	Candidate. Tool depends on the repository host
Security	Encrypted connections, encryption at rest, secrets kept out of the repository, role-based access	Body photos are sensitive	Decided
Operations	Monitoring, alerts, daily backups	A pilot with real shoppers	Candidate
Devices	One in-store screen for the mirror, one staff tablet	Pilot shop only	Open (OPS-02, sprint 4)
Kiabi systems	Catalog and size charts, e-commerce cart, loyalty card with NFC tag	Owned by Kiabi. Access via the Kiabi adapter	To obtain from Kiabi (OPS-01)

5. Players

Player	Role
Shopper	Creates an avatar, tries items on, buys
Store associate	Receives size requests on the tablet and brings the item
Kiabi e-commerce and marketing team	Reads the dashboard, owns catalog and cart
Kiabi IT and data	Gives access to catalog, size charts, cart and loyalty systems
Product team	Elie Zaylaa, Boyu Fu, Muhammad Rao, Hadeed: build and run the platform
Avatar technology provider	Only if sprint 1 decides to buy (SP-01)
School jury	Reviews each checkpoint on Gandalf

6. Data flows

Step	Flow	Category touched
1	The shopper consents and sends photos from the app. The API passes them to the avatar service, which returns the avatar. Photos are deleted afterwards	Functional, Application, Techniques
2	The avatar, the confirmed height and size and the consent log are stored	Application, Systems
3	The Kiabi adapter pulls catalog items and size charts. The fit engine compares the avatar with each size chart	Application, Techniques
4	The app shows the garment on the avatar and the fit per zone. The shopper picks a size and the adapter hands the article to Kiabi's cart	Functional, Application
5	In store, a QR code (later the NFC tap) identifies the customer and the mirror loads the same avatar	Functional, Systems
6	A size request goes from the mirror through the API to the staff tablet	Application
7	Every step sends an event to analytics. The dashboard reads aggregates only	Application, Systems

7. Environments

Environment	Purpose	Data
Local	Development on each laptop	Fake photos, sample catalog
Dev	Shared integration, deployed on every merge	Synthetic data
Staging	Sprint reviews and demos, same set-up as production	Test accounts, sample Kiabi feed
Production	Pilot with real shoppers on the web and in one shop	Real data, EU hosting, backups, consent log

8. Open technical decisions

Decision	Settled by	Date

Build or buy the avatar pipeline	SP-01	End of sprint 1, Fri 2026-11-20
Hosting provider	Muhammad, with the team	Sprint 1
Delivery pipeline tool	Muhammad	Sprint 1
Kiabi access method (API or file feed)	OPS-01 answer from Kiabi	Sprint 1 to 2
Mirror and tablet hardware	OPS-02	Sprint 4

All decisions are recorded in `C-backlog/decision-log.docx`.